THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** copper	Best conductor.
2	**(d)** lower than over sea	Warm air rises, low pressure.
3	**(c)** white loose cotton	Reflect and breathe.
4	**(b)** radiation	Air is a poor conductor.
5	**(b)** sea	Sea is warmer at night.
6	**(b)** layers of woollen clothes	Trapped air.
7	**(b)** convection	Hot air rises.
8	**(d)** water takes longer to heat	High heat capacity.
9	**(b)** air gets trapped between layers	Insulation.
10	**(a)** radiation	Sideways transfer.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Metals are good conductors of heat (1); Wood is an insulator (1).	1+1
12	Copper is a good conductor (1); Even and fast heating (1).	1+1
13	Heat flows from the fish to the ice (1); The fish stays cold (1).	1+1
14	Hot air inside is lighter than the surrounding air (1); Convection lifts it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Radiation to nearby objects (1); Convection moves warm air (1); Conduction in the heating element (1).	1+1+1
16	Metal conducts (1); Plastic insulates (1); Wool reduces heat loss (1).	1+1+1
17	Hot coal radiates (1); Food absorbs (1); Convection also helps (1).	1+1+1
	OR: Land is hottest then (1); Larger temperature difference (1); Strong convection (1).	1+1+1
18	Heat conducts to both ends (1); Equal for equal distance (1); Wax pins fall from the centre outward (1).	1+1+1
	OR: Conduction through the mug (1); Ceramics conduct slowly (1); Metal would be worse (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Conduction: wax pins on rod (1); Convection: permanganate crystals in water (1); Radiation: hand near a lamp (1); Observations noted (1); Conclusions drawn (1).	1+1+1+1+1
	OR: Black pipes absorb (1); Glass cover traps heat (1); Water circulates by convection (1); Hot water tank insulated (1); Saves fuel (1).	1+1+1+1+1
20	Vacuum prevents conduction and convection (1); Silvered walls prevent radiation (1); Loss of vacuum lets air carry heat (1); Contents lose heat faster (1); Flask useless (1).	1+1+1+1+1
	OR: Metals conduct (1); Lose heat quickly (1); Earthen pots conduct less (1); Keep warm longer (1); Matka keeps water cool (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
- Writing that conduction needs the movement of particles from place to place; in conduction the particles only pass on the heat and **stay in position**.
- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
- Thinking that heat from the Sun reaches us by conduction or convection; it is **radiation**, which needs no medium.
- Writing that dark colours are better in summer; **light** colours reflect more radiation.
- Confusing heat with temperature; temperature is a measure of hotness, heat is energy in transfer.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** water cools slowly	Takes more time to cool.
2	**(c)** reflect sun and let air circulate	Keep cool.
3	**(a)** black absorbs more radiation	Absorption.
4	**(c)** warm air rises and carries it	Thermals.
5	**(d)** reflect heat	Reduce heat gain.
6	**(d)** conduction	Needs particles.
7	**(a)** hot gases rise	Convection.
8	**(c)** reflect sunlight	Keep cool.
9	**(b)** evaporation	It is a change of state.
10	**(a)** convection spreads heat through the water	Convection.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reflects sunlight (1); Keeps the house cool (1).	1+1
12	Soot comes from the flame (1); It absorbs heat better but conducts poorly (1).	1+1
13	Radiation reaches us sideways (1); Hot gases rise above (1).	1+1
14	Air layer between them (1); Air is an insulator (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Two identical cans painted black and white (1); Fill with water and place in sun (1); Black one is hotter after some time (1).	1+1+1
16	Radiation passes through plastic (1); Water absorbs (1); Convection spreads heat (1).	1+1+1
17	Poor conductors (1); Less heat loss (1); Metal loses quickly (1).	1+1+1
	OR: Reflector concentrates sunlight (1); Black surface absorbs (1); Glass lid traps heat (1).	1+1+1
18	Air moves (1); Sweat evaporates (1); Heat removed (1).	1+1+1
	OR: Moves warm air (1); Evaporation cools (1); Brings cool air (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Day: land hotter (1); Air rises over land (1); Cool air from sea (1); Night: reverse (1); Cool breezes (1).	1+1+1+1+1
	OR: Heat conducts along the rod (1); Hand end becomes hot (1); Use insulating gloves (1); Tongs (1); Avoid burns (1).	1+1+1+1+1
20	Light colours in summer (1); Dark in winter (1); Cotton breathes (1); Wool insulates (1); White walls reflect (1).	1+1+1+1+1
	OR: White roofs (1); Thick walls (1); Ventilation (1); Shades (1); Trees (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
- Writing that conduction needs the movement of particles from place to place; in conduction the particles only pass on the heat and **stay in position**.
- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** good emitter of heat	Dull black emits better.
2	**(c)** ice wrapped in thick wool	Wool insulates.
3	**(d)** near the ceiling	Warm air rises.
4	**(c)** reduces radiation and holds heat	Shiny surface.
5	**(d)** metal is a better conductor	Conduction.
6	**(d)** cold air sinks and cools the room	Convection.
7	**(a)** to reduce heat loss	Saves fuel.
8	**(b)** vacuum prevents conduction and convection	No medium.
9	**(c)** cold air sinks to cool below	Convection.
10	**(b)** thermocol is an insulator	Reduces heat flow.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reflect radiation (1); Keep contents cold (1).	1+1
12	Sea air is warmer and rises (1); Cooler land air moves in (1).	1+1
13	Reduce heat loss (1); Conserve body heat (1).	1+1
14	Poor conductors (1); Heat takes time to pass (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Air gets trapped (1); Adjustable (1); Better insulation (1).	1+1+1
16	Thermocol or foam (1); Prevent heat entering (1); Save electricity (1).	1+1+1
17	A shiny surface radiates less heat (1); A dull black surface radiates more heat (1); So the shiny kettle loses heat slowly (1).	1+1+1
	OR: Water is a poor conductor (1); Warm water stays on top (1); Little convection downward (1).	1+1+1
18	Lake cools slowly (1); Breeze comes (1); Convection (1).	1+1+1
	OR: No direct radiation (1); Less absorption (1); Air circulates (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solids: conduction (1); Liquids: convection (1); Gases: convection (1); Radiation in all (1); Examples for each (1).	1+1+1+1+1
	OR: Radiation ignites nearby trees (1); Convection carries hot gases (1); Wind helps (1); Embers carried (1); Firebreaks stop (1).	1+1+1+1+1
20	Geyser: element heats water (1); Convection circulates (1); Iron: plate conducts (1); Fridge: insulation (1); Freezer at top (1).	1+1+1+1+1
	OR: Igloo: snow traps air (1); Mud house: poor conductor (1); Hill house: thick walls (1); Roofs and windows (1); Heat retained (1).	1+1+1+1+1

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